

Veeranjan Reddy Birasam

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Summary

Software Engineer with 3+ years of experience designing and deploying scalable backend systems, APIs, and microservices using Python (Flask, Django, FastAPI). Skilled in PostgreSQL, MongoDB, Redis, and SQLAlchemy for data-driven architectures with optimized performance. Proficient in containerization, CI/CD automation, and cloud-native deployments leveraging AWS, Azure, Docker, and Kubernetes. Strong expertise in TDD, secure coding, and API security, consistently improving application reliability, scalability, and system efficiency while driving measurable impact across enterprise-grade environments.

Technical Skills

Programming & Application Development: Python (Flask, Django, FastAPI), Java, JavaScript, SQL, C++, Shell Scripting, Object-Oriented Programming (OOP), REST APIs, GraphQL, NumPy, Pandas, Matplotlib, PyTest, Unit Testing

Backend & Databases: PostgreSQL, MySQL, MongoDB, SQLite, Redis, Cassandra, SQLAlchemy, Data Modeling, Query Optimization, Stored Procedures, Caching Strategies, Database Design

Web Development & Frontend: HTML5, CSS3, JavaScript (React.js), Jinja2 Templates, Responsive UI, WebSockets, Authentication (JWT, OAuth 2.0), MVC Architecture

Cloud Computing & DevOps: Amazon AWS (EC2, S3, Lambda, RDS), Microsoft Azure (App Service, Blob Storage), Docker, Kubernetes, Terraform, CI/CD Pipelines (GitHub Actions, Jenkins), Linux/Unix, Role-Based Access Control (RBAC)

Software Engineering, Optimization & Security: Agile (Scrum/Kanban), Design Patterns, Test-Driven Development (TDD), Version Control (Git/GitHub), Code Review, Logging & Monitoring, Debugging, Secure Coding (OWASP), API Security

Professional Experience

Software Engineer, *Databricks*

Aug 2024 – Present | Remote, USA

- Developed enterprise-grade backend services using Python and Django Rest Framework, enabling secure multi-tenant APIs, processing 1.2M+ daily requests, and reducing response latency by 33% across distributed environments.
- Designed PostgreSQL schemas with advanced indexing and query optimization, boosting performance by 37% and eliminating critical data retrieval bottlenecks across enterprise-scale reporting systems.
- Implemented OAuth 2.0 and JWT-based authentication modules, improving role-based access control, reducing security incidents by 41%, and ensuring compliance with strict enterprise security standards.
- Automated containerized deployments with Docker, GitHub Actions, and Terraform on AWS, accelerating release cycles by 35% and enabling seamless rollback strategies for cloud-native applications.
- Built PyTest-driven unit and integration testing frameworks, achieving 91% coverage, reducing post-release defects by 29%, and strengthening reliability of backend microservices in production.
- Integrated observability with Datadog and AWS CloudWatch, detecting anomalies proactively, reducing downtime by 34%, and improving real-time visibility into distributed backend service operations.

Software Engineer, *LTIMindtree*

Mar 2021 – Jul 2023 | Remote, India

- Built Python Flask-based backend services handling 120,000+ API requests daily, reducing latency by 31%, improving authentication workflows, and enabling faster response times across client-facing web applications.
- Designed PostgreSQL database schemas with SQLAlchemy ORM, implementing query optimization and indexing techniques that boosted reporting efficiency by 36% and reduced transaction errors in production workloads.
- Developed role-based access modules using JWT and OAuth 2.0, strengthening authentication, increasing compliance, and reducing unauthorized access attempts by 43% across distributed microservice architectures.
- Automated CI/CD pipelines with Jenkins and Docker containers, improving deployment frequency by 28%, minimizing manual intervention, and ensuring consistent environment provisioning across multiple staging and production systems.
- Applied Test-Driven Development practices with PyTest and Unit Testing, increasing coverage to 88%, reducing regression bugs by 27%, and enhancing reliability across high-priority enterprise applications.
- Configured Redis caching and implemented database partitioning strategies, enhancing system throughput by 32% and enabling scalable backend performance during peak traffic spikes from 15,000+ concurrent users.
- Integrated structured logging and monitoring with ELK Stack, reducing incident resolution times by 35% and enabling proactive alerting for critical service outages across enterprise platforms.
- Designed responsive web interfaces with Jinja2 templates and WebSocket integration, enabling real-time dashboards that improved decision-making accuracy by 40% for business stakeholders.

Education

Master of Science, *University of Illinois - Springfield*
Management Information Systems

Aug 2023 – Dec 2024 | Illinois, USA

Bachelor of Engineering, *New Horizon College of Engineering*

Aug 2018 – Jul 2022 | Karnataka, India